

Problem 12

When are the following equalities possible:

$$1^\circ A + B = \bar{A}; \quad 2^\circ AB = \bar{A}; \quad 3^\circ A + B = AB?$$

Solution

We will analyze each equality to determine when it can hold.

Part 1°: When is $A + B = \bar{A}$?

Analysis

The equation $A + B = \bar{A}$ means $A \cup B = A^c$.

Step 1: Take Complement of Both Sides

$$\overline{A + B} = \overline{\bar{A}}$$

$$\bar{A}\bar{B} = A$$

So we need: $A = \bar{A}\bar{B}$

Step 2: Substitute Back

From $A = \bar{A}\bar{B}$, we can substitute into the original equation:

$$\bar{A}\bar{B} + B = \bar{A}$$

Using distributive law in reverse:

$$\bar{A}\bar{B} + B = B(\bar{A} + 1) = B \cdot S = B$$

Wait, let me reconsider. Using the identity $\bar{A}\bar{B} + B$:

$$\begin{aligned} \bar{A}\bar{B} + B &= \bar{A}\bar{B} + B(1) = \bar{A}\bar{B} + B(\bar{A} + A) \\ &= \bar{A}\bar{B} + B\bar{A} + BA = \bar{A}(\bar{B} + B) + AB = \bar{A} + AB \end{aligned}$$

So we need: $\bar{A} + AB = \bar{A}$

This is satisfied by absorption if $AB \subseteq \bar{A}$, i.e., $AB = \emptyset$.

Step 3: Conclusion

From $A = \bar{A}\bar{B}$:

Multiply both sides by A :

$$A = A\bar{A}\bar{B} = \emptyset$$

So $A = \emptyset$ (the impossible event).

Let's verify: If $A = \emptyset$, then:

$$A + B = \emptyset + B = B$$

$$\bar{A} = \bar{\emptyset} = S$$

For the equation to hold: $B = S$ (the certain event).

Answer:

$$\boxed{A = \emptyset \text{ and } B = S}$$

Part 2°: When is $AB = \bar{A}$?

Analysis

The equation $AB = \bar{A}$ means $A \cap B = A^c$.

Step 1: Intersect Both Sides with A

$$A \cdot AB = A\bar{A}$$

$$AB = \emptyset$$

But we also have $AB = \bar{A}$, so:

$$\bar{A} = \emptyset$$

$$A = S$$

Step 2: Check Consistency

If $A = S$, then:

$$AB = SB = B$$

$$\bar{A} = \bar{S} = \emptyset$$

For the equation to hold: $B = \emptyset$.

Answer:

$$\boxed{A = S \text{ and } B = \emptyset}$$

Part 3°: When is $A + B = AB$?

Analysis

The equation $A + B = AB$ means $A \cup B = A \cap B$.

This happens if and only if $A = B$.

Proof

Direction 1: If $A = B$, then:

$$A + B = A + A = A$$

$$AB = AA = A$$

So $A + B = AB$.

Direction 2: If $A + B = AB$, we need to show $A = B$.

From $A + B = AB$:

- Since $AB \subseteq A + B$ always, and we have $A + B = AB$, this is consistent.
- Also, $A \subseteq A + B$ and $B \subseteq A + B$ always hold.

Since $A + B = AB$:

$$A \subseteq AB \text{ and } B \subseteq AB$$

From $A \subseteq AB$: Every element in A is also in $AB = A \cap B$, so every element in A is in B . Thus $A \subseteq B$.

From $B \subseteq AB$: Every element in B is also in $AB = A \cap B$, so every element in B is in A . Thus $B \subseteq A$.

Therefore, $A = B$.

Answer:

$$\boxed{A = B}$$

Verification of Results

Verify Part 1°: $A = \emptyset, B = S$

$$A + B = \emptyset + S = S$$

$$\bar{A} = \bar{\emptyset} = S$$

Indeed, $A + B = \bar{A}$.

Verify Part 2°: $A = S, B = \emptyset$

$$AB = S \cdot \emptyset = \emptyset$$

$$\bar{A} = \bar{S} = \emptyset$$

Indeed, $AB = \bar{A}$.

Verify Part 3°: $A = B$

$$A + B = A + A = A$$

$$AB = A \cdot A = A$$

Indeed, $A + B = AB$.

Probabilistic Interpretation

Part 1°: A never occurs (impossible event), and B always occurs (certain event).

Part 2°: A always occurs (certain event), and B never occurs (impossible event).

Part 3°: A and B are the same event.

Final Answer

1°	$A + B = \bar{A} \iff A = \emptyset \text{ and } B = S$
2°	$AB = \bar{A} \iff A = S \text{ and } B = \emptyset$
3°	$A + B = AB \iff A = B$

Summary:

- The first equality requires A to be impossible and B to be certain
- The second equality requires A to be certain and B to be impossible
- The third equality requires A and B to be identical events

Key insight: These are very restrictive conditions. The first two require extreme cases (impossible or certain events), while the third requires complete equality of the events.